Formal Language and Automata Theory

FLAT

Topics Covered

1. Introduction to flat
2. Applications of flat
3. Real world example
4. Importance of flat

Introduction to flat

- FLAT is a theortical branch of computer science that deals with:
- -Formal languages
- -Grammer
- -Automata(machines)
- it studies how machines compute and recognize parrerns

Types of Automata

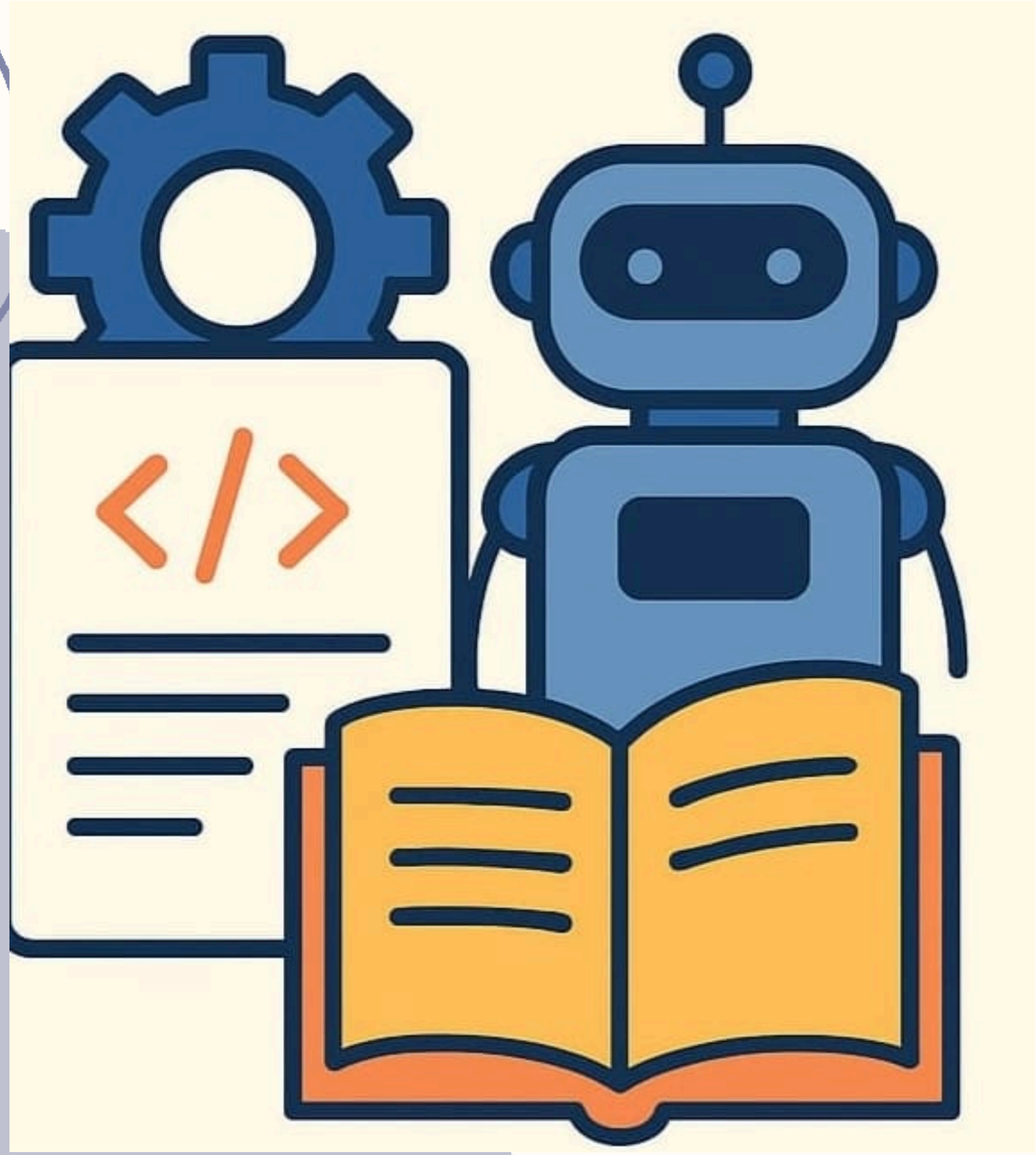
- Finite Automata
- Pushdown automata
- Turning machines

Applications of flat

- compiler design
- Artifical intelligence
- Cyber security
- Digital circuit design

Real world example

- Search Engines: pattern matching
- Lexical Analyzers: compilers
- Text Editor: syntax highlighting
- AI: natural language generation



Importance of flat

- Helps in designing parser and compliers
- Flat is essential in understanding how compilers work
- Improves problem solving
- Applications in security &protocols

Flat in daily life

1. Google search and text predictions
2. Programming languages
3. Voice assistants(like Alexa,siri)
4. Chatbots and AI assistants

Difference between DFA and NFA

Deterministic Finite Automata

- One path, no confusion, strict machine. ✅ Used in real application
- Easier to implement in hardware or software.

Nondeterministic Finite Automata

- Many possible paths, flexible machine. 🧪 Great for design & theory
- From any state, the machine can move to zero or more next states for a given input.

Thank you!